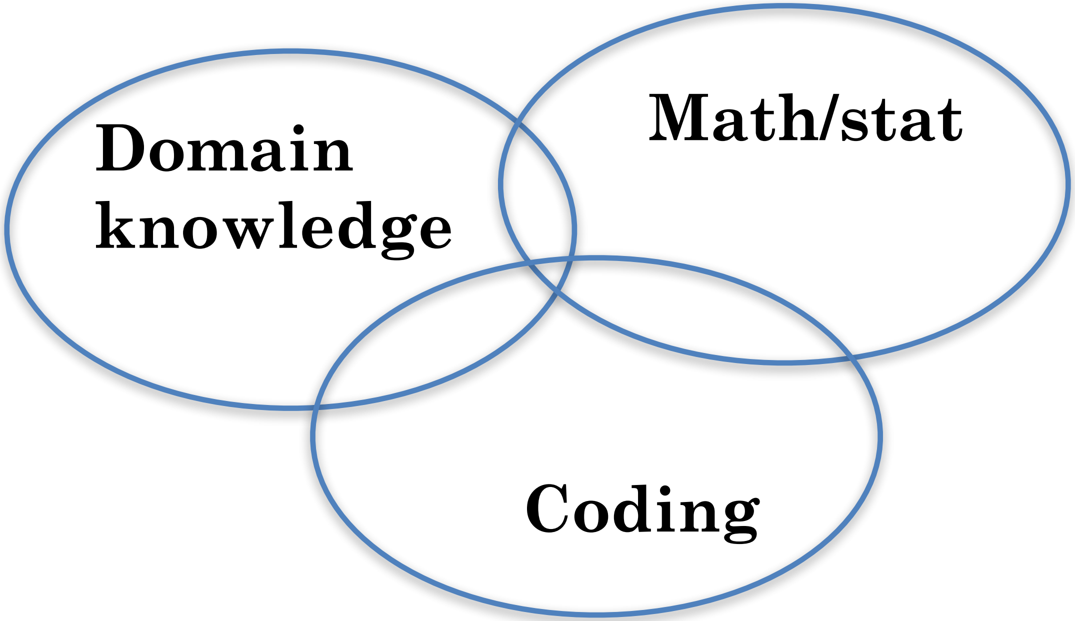
Introduction to Probability and Statistics Fourteenth Edition

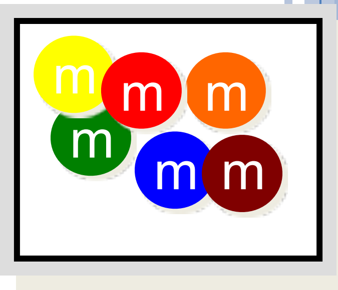
Chapter 14
Analysis of Categorical Data

14.1 the Multinomial experiment and the Chi-square statistic

Data science/AI



Introduction



- Many experiments result in measurements that are **qualitative** or **categorical** rather than quantitative.
  - People classified by ethnic origin
  - Cars classified by color
  - M&M®s classified by type (plain or peanut)
- These data sets have the characteristics of a **multinomial experiment**.

Binomial Experiment=Sum of n Bernoulli trials

1. The experiment consists of ***n* identical Bernoulli trials**.
2. Each trial results in **one of 2 categories (category 1 = success, category 2 = failure)**.
3. **The probability of success is $p_1 = p$, the probabiltiy of failure is $p_2 = (1 - p)$.**
4. The trials are **independent**.
5. We are interested in the number of outcomes in each category, $\boldsymbol{O_{1,}}$ $\boldsymbol{O_2}$ with $\boldsymbol{O_1 + O_2 = n}$.

An Example

- $n = 5$, suppose we toss a fair coin 5 times.

Outcome: (0, 1, 0, 1, 1)

$X \sim$ Binomial(5,0.5).

X = 3.

$P(X = x) = C_x^n p^x (1 - p)^{(n-x)}, x = 0,1,\ldots,n.$

- $(O_1, O_2) \sim$ Multinomial($n = 5, p_1 = 0.5, p_2 = 0.5$)

$(O_1, O_2) = (3,2)$

$P(O_1 = o_1, O_2 = o_2) = C^n_{n_1,n_2} p_1^{o_1} p_2^{o_2},\ C^n_{n_1,n_2} = \frac{n!}{n_1!n_2!}$

Comparisons: ch 9 versus ch 14

- $\begin{cases} H_0 : p = p_0 \\ H_1 : p \neq p_0 \end{cases}$

Estimator $\hat{p} = \dfrac{x}{n}$

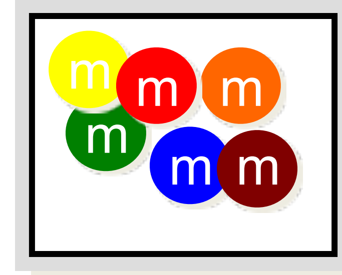
Test statistic: $Z_{stat} = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1 - p_9)}} \sim Z$

- $\begin{cases} H_0 : p_1 = p_{10} = p_{20}, \ldots, p_k = p_{k0} \\ H_1 : \text{At least one equality in } H_0 \text{ does not hold} \end{cases}$

Estimator $\hat{p}_i = \dfrac{O_i}{n}$

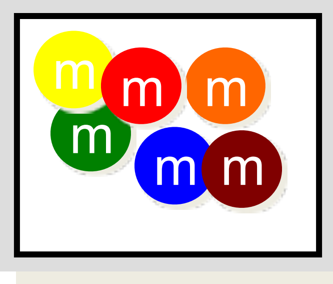
Test statistic: $\displaystyle\sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{df}, \; E_i = p_{i0} \times n$

The Multinomial Experiment



1. The experiment consists of ***n* identical trials**.
2. Each trial results in **one of *k* categories**.
3. The probability that the outcome falls into a particular category i on a single trial is p_i and **remains constant** from trial to trial. The sum of all k probabilities, $\boldsymbol{p_1+p_2+\ldots+p_k=1}$.
4. The trials are **independent**.
5. We are interested in the number of outcomes in each category, $\boldsymbol{O_1, O_2, \ldots O_k}$ with $\boldsymbol{O_1+O_2+\ldots+O_k=n}$.

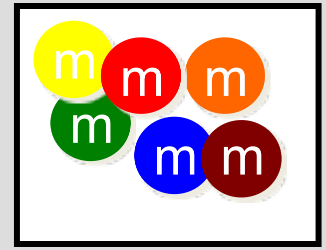
The Binomial Experiment



- A special case of the multinomial experiment with $k = 2$.
- Categories 1 and 2: **success and failure**
- p_1 and p_2: ***p* and *q***
- O_1 and O_2: ***x* and *n-x***
- We made inferences about p (and $q = 1 - p$)

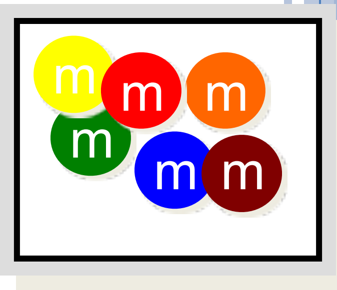
In the multinomial experiment, we make inferences about all the probabilities, p_1, p_2, p_3 …p_k.

Pearson's Chi-Square Statistic



- We have some preconceived idea about the values of the p_i and want to use sample information to see if we are correct.
- The **expected number** of times that outcome i will occur is $\boldsymbol{E_i = np_{i0}}$.
- If the **observed cell counts, $\boldsymbol{O_i}$**, are too far from what we hypothesize under H_0, the more likely it is that H_0 should be rejected.

PEARSON'S CHI-SQUARE STATISTIC



- We use the Pearson chi-square statistic:

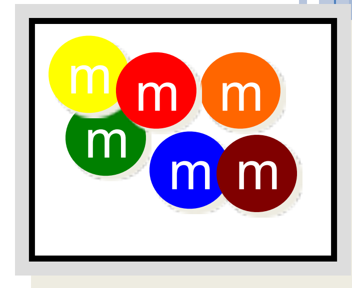
$$X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- When H_0 is true, the differences *O-E* will be small, but large when H_0 is false.
- Look for **large values of X^2** based on the chi-square distribution with a particular number of degrees of freedom.

Intuition but Not rigorous:

- $$\left(\frac{\hat{p}-p_0}{\sqrt{p_0 q_0/n}}\right)^2=\left(\frac{n\hat{p}-np_0}{\sqrt{np_0 q_0}}\right)^2=\frac{(O-E)^2}{Eq_0}\approx\frac{(O-E)^2}{E}$$

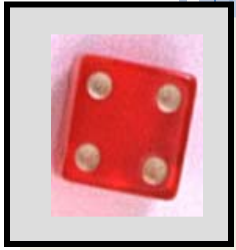
DEGREES OF FREEDOM



- These will be different depending on the application.

1. Start with the number of categories or cells in the experiment.
2. Subtract 1*df* for each linear restriction on the cell probabilities. (You always lose 1 *df* since $p_1 + p_2 + \ldots + p_k = 1$.)
3. Subtract 1 *df* for every population parameter you have to estimate to calculate or estimate E_i.

14.2 Testing Specified Cell Probabilities: the Goodness-of-Fit Test



THE GOODNESS OF FIT TEST

- The simplest of the applications.
- A single categorical variable is measured, and exact numerical values are specified for each of the p_i.
- Expected cell counts are $E_i = np_{i0}$
- Degrees of freedom: $df = k-1$

$$\text{Test statistic: } X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Example

- Toss a die 300 times with the following results. Is the die fair or biased?

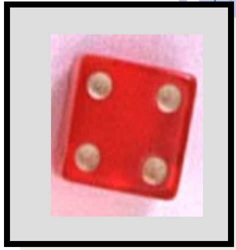
Upper Face	1	2	3	4	5	6
Number of times	50	39	45	62	61	43

- A multinomial experiment with $k = 6$ and O_1 to O_6 given in the table.
- We test:

H_0: $p_1 = 1/6$; $p_2 = 1/6$;…$p_6 = 1/6$ (die is fair)

H_a: at least one p_i is different from 1/6 (die is biased)

Example



- Calculate the expected cell counts:

$E_i = np_i = 300(1/6) = 50$

Upper Face	1	2	3	4	5	6
O_i	50	39	45	62	61	43
E_i	50	50	50	50	50	50

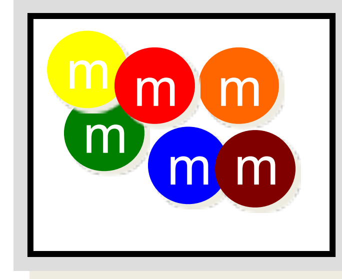
- Test statistic and rejection region:

$$X^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(50-50)^2}{50} + \frac{(39-50)^2}{50} + \ldots + \frac{(43-50)^2}{50} = 9.2$$

$$\text{Reject}\,H_0 \text{ if } X^2 > \chi^2_{.05} = 11.07 \text{ with } k-1 = 6-1 = 5 \text{ df.}$$

Do not reject H_0. There is insufficient evidence to indicate that the die is biased.

Some Notes



- The test statistic, X^2 has only an approximate chi-square distribution.
- For the approximation to be accurate, statisticians recommend $E_i \geq 5$ for all cells.
- Goodness of fit tests are different from previous tests since the experimenter uses H_0 for the model he thinks is *true*.

H_0: model is correct (as specified)
H_a: model is not correct

- Be careful not to accept H_0 (say the model is correct) without reporting β.

14.3 Contingency tables: A two-Way Classification

Contingency Tables: A Two-Way Classification

- The experimenter measures **two qualitative variables** to generate **bivariate data**.
  - Gender and colorblindness
  - Age and opinion
  - Professorial rank and type of university
- Summarize the data by counting the **observed** number of outcomes in each of the intersections of category levels in a **contingency table**.

Raw data

Index	Factor A (A, B, C, D)	Factor B (1, 2)
1	A	1
2	B	2
3	C	1
4	D	1
5	A	2
6	A	2
7	B	1
8	B	2

Contigency Table

	A	B	C	D
1	1	1	1	1
2	2	2	0	0

Recall the definition of "Independence" of two events

- If $P(A) = 0.5$, $P(B) = 0.6$, $P(A \cap B) = 0.3$. Are A and B are indepdent?

"Independence" of two events

- Originally, if events A and B are indepedent, if and only if $P(A|B) = P(A)$ and $P(B|A) = P(B)$.

Because $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$.

If A and B are indepedent $P(A|B) = P(A)$ equals

$\dfrac{P(A \cap B)}{P(B)} = P(A)$ and thus $P(A \cap B) = P(A)P(B)$.

Similarly, $P(B|A) = \dfrac{P(B \cap A)}{P(A)}$. $P(B|A) = P(B)$ equals

$\dfrac{P(B \cap A)}{P(A)} = P(B)$, and thus $P(A \cap B) = P(A)P(B)$.

Therefore, a second definition for events A and B being indepedent, if and only if $P(A \cap B) = P(A)P(B)$.

R X *C* CONTINGENCY TABLE

- The contingency table has *r* rows and *c* columns—*rc* total cells.

	1	2	…	*c*
1	O_{11}	O_{12}	…	O_{1c}
2	O_{21}	O_{22}	…	O_{2c}
…	…	…	…	….
r	O_{r1}	O_{r2}	…	O_{rc}

- We study the relationship between the two variables. Is one method of classification **contingent** or **dependent** on the other?

Does the distribution of measurements in the various categories for variable 1 depend on which category of variable 2 is being observed?
If not, the variables are independent.

p_{ij}, p_i, p_j

- ✓ p_{ij} = *P(falling in row i and column j)*
- ✓ p_i= P(falling in row *i*)
- ✓ p_j = P(falling in column *j*)

	1	2	…		…	c	
1	O_{11}	O_{12}	…		…	O_{1c}	r_1
2	O_{21}	O_{22}	…		…	O_{2c}	r_2
…	…	…	…		…	….	
							r_i
r	O_{r1}	O_{r2}	…		…	O_{rc}	
	c_1	c_2		c_j			n

Chi-Square Test of Independence

H_0: classifications are independent

H_a: classifications are dependent

H_0: $p_{ij} = p_i p_j$ for all i, j

H_a: At least one probability from above is different from the specified values

- Observed cell counts are ***O_{ij}*** for row *i* and column *j*.
- Expected cell counts are ***$E_{ij} = np_{ij}$***
  - ✓ If H_0 is true and the classifications are independent,
  - ✓ $p_{ij} = p_i p_j$ = P(falling in row *i*)P(falling in row *j*)

Chi-Square Test of Independence

Estimate marginal probabilities:

$$\hat{p}_i = \frac{r_i}{n} \qquad \hat{p}_j = \frac{c_j}{n}$$

Test statistic: $X^2 = \sum \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}$

If H_0 is true

$$\hat{E}_{ij} = n\hat{p}_{ij} = n\hat{p}_i\hat{p}_j = n\frac{r_i}{n}\frac{c_j}{n} = \frac{r_i c_j}{n}$$

- The test statistic has an approximate chi-square distribution with ***df* = (*r*-1)(*c*-1).**

EXAMPLE

Suppose we would like to test independence for classifcations.

A\B O_{ij}	**1**	**2**	**Row sum**
1	4	6	10
2	5	5	10
Column sum	**9**	**11**	**20**

A\B $\hat{E}_{ij}$	**1**	**2**	**Row sum**
1	90/20	110/20	10
2	90/20	110/20	10
Column sum	**9**	**11**	**20**

EXAMPLE

Furniture defects are classified according to type of defect and shift on which it was made.

	Shift			
Type	**1**	**2**	**3**	***Total***
A	15	26	33	**74**
B	21	31	17	**69**
C	45	34	49	**128**
D	13	5	20	**38**
Total	**94**	**96**	**119**	**309**

Do the data present sufficient evidence to indicate that the type of furniture defect varies with the shift during which the piece of furniture is produced? Test at the 1% level of significance.

H_0: type of defect is independent of shift

H_a: type of defect depends on the shift

The Furniture Problem

- Calculate the expected cell counts. For example:

$$\hat{E}_{12} = \frac{r_1 c_2}{n} = \frac{74(96)}{309} = 22.99$$

MY APPLET

```
Chi-Square Test: 1, 2, 3
Expected counts are printed below o
Chi-Square contributions are printe
            1        2        3   Total
    1      15       26       33      74
        22.51    22.99    28.50
        2.506    0.394    0.711

    2      21       31       17      69
        20.99    21.44    26.57
        0.000    4.266    3.449
```

```
Chi-Sq = 19.178, DF = 6, P-
Value = 0.004
```

Reject H_0. There is sufficient evidence to indicate that the proportion of defect types vary from shift to shift.

$$\text{Test statistic: } X^2 = \sum \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} =$$

$$\frac{(15-22.51)^2}{22.51} + \frac{(26-22.99)^2}{22.99} + \ldots + \frac{(\ }{14.63}$$

$$\text{Reject } H_0 \text{ if } X^2 > \chi^2_{.05} = 11.07 \text{ with } (r-1)(c-1) = 6 \text{ df.}$$

14.4 Comparing several Multinomial populations: A two-Way Classification with Fixed row or Column totals

Comparing Multinomial Populations

- Sometimes researchers design an experiment so that the number of experimental units falling in one set of categories is fixed in advance.
- **Example:** An experimenter selects 900 patients who have been treated for flu prevention. She selects 300 from each of three types—no vaccine, one shot, and two shots.

The column totals have been fixed in advance!

	No Vaccine	One Shot	Two Shots	*Total*
Flu				r_1
No Flu				r_2
Total	300	300	300	n = 900

Comparing Multinomial Populations

- Each of the *c* columns (or *r* rows) whose totals been fixed in advance is actually a single multinomial experiment.
- The chi-square test of independence with (r-1)(c-1) *df* is **equivalent to** a test of the equality of *c* (or *r*) multinomial populations.

✓Three binomial populations—no vaccine, one shot and two shots.
✓Is the probability of getting the flu independent of the type of flu prevention used?

	No Vaccine	**One Shot**	**Two Shots**	***Total***
Flu				r_1
No Flu				r_2
Total	**300**	**300**	**300**	n **= 900**

EXAMPLE

Random samples of 200 voters in each of four wards were surveyed and asked if they favor candidate A in a local election.

MY APPLET

	Ward				
	1	**2**	**3**	**4**	***Total***
Favor A	76	53	59	48	**236**
Do not favor A	124	147	141	152	**564**
Total	**200**	**200**	**200**	**200**	**800**

Do the data present sufficient evidence to indicate that the the fraction of voters favoring candidate A differs in the four wards?

H_0: fraction favoring A is independent of ward

H_a: fraction favoring A depends on the ward

H_0: $p_1 = p_2 = p_3 = p_4$

where p_i = fraction favoring A in each of the four wards

O_{ij}	Ward				
	1	2	3	4	*Total*
Favor A	76	53	59	48	**236**
Do not favor A	124	147	141	152	**564**
Total	**200**	**200**	**200**	**200**	**800**

236/800*200= 59

564/800*200= 141

$\hat{E}_{ij}$	Ward				
	1	2	3	4	*Total*
Favor A	59	59	59	59	**236**
Do not favor A	141	141	141	141	**564**
Total	**200**	**200**	**200**	**200**	**800**

The Voter Problem

MY APPLET

- Calculate the expected cell counts. For example:

$$\hat{E}_{12} = \frac{r_1 c_2}{n} = \frac{236(200)}{800} = 59$$

$$\text{Test statistic}: X^2 = \sum \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} =$$

$$\frac{(76-59)^2}{59} + \frac{(53-59)^2}{59} + \ldots + \frac{(152-141)^2}{141} = 10.722$$

$$\text{Reject } \mathrm{H}_0 \text{ if } \mathrm{X}^2 > \chi^2_{.05} = 7.81 \text{ with } (r-1)(c-$$

Reject H_0. There is sufficient evidence to indicate that the fraction of voters favoring A varies from ward to ward.

MINITAB OUTPUT


```
Chi-Square Test: 1, 2, 3, 4

Expected counts are printed below observed counts
Chi-Square contributions are printed below expected counts

            1       2       3       4  Total
    1      76      53      59      48    236
        59.00   59.00   59.00   59.00
        4.898   0.610   0.000   2.051

    2     124     147     141     152    564
       141.00  141.00  141.00  141.00
        2.050   0.255   0.000   0.858

Total     200     200     200     200    800
```

```
Chi-Sq = 10.722, DF = 3, P-Value = 0.013
```

MY APPLET

The Voter Problem

Since we know that there are differences among the four wards, what are the nature of the differences?

Look at the proportions in favor of candidate A in the four wards.

Ward	1	2	3	4
Favor A	76/200=.38	53/200 = .27	59/200 = .30	48/200 = .24

Candidate A is doing best in the first ward, and worst in the fourth ward. More importantly, he does not have a majority of the vote in any of the wards!

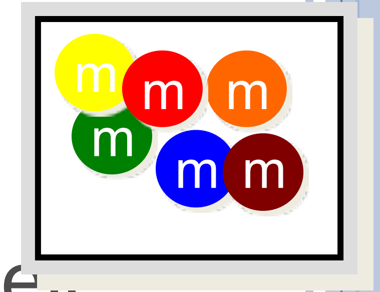
14.5 Other topics in Categorical Data Analysis

Assumptions

Assumptions for Pearson's Chi-Square:

1. The cell counts O_1, O_2, …,O_k must satisfy the conditions of a multinomial experiment, or a set of multinomial experiments created by fixing either the row or the column totals.
2. The expected cell counts E_1, E_2, …, E_k should equal or exceed five.

Assumptions



- ✓ When you calculate the expected cell counts, if you find that one or more is less than five, these options are available to you:

1. **Choose a larger sample size *n***. The larger the sample size, the closer the chi-square distribution will approximate the distribution of your test statistic X^2.
2. It may be possible to **combine one or more of the cells** with small expected cell counts, thereby satisfying the assumption.

Key Concepts

I. The Multinomial Experiment

1. There are n identical trials, and each outcome falls into one of k categories.
2. The probability of falling into category i is p_i and remains constant from trial to trial.
3. The trials are independent, $\Sigma\, p_i = 1$, and we measure O_i, the number of observations that fall into each of the k categories.

II. Pearson's Chi-Square Statistic

$$X^2 = \sum \frac{(O_i - E_i)^2}{E_i} \qquad \text{where } E_i = np_i$$

which has an approximate chi-square distribution with **degrees of freedom** determined by the application.

Key Concepts

III. The Goodness-of-Fit Test

1. This is a one-way classification with cell probabilities specified in H_0.
2. Use the chi-square statistic with $E_i = np_i$ calculated with the hypothesized probabilities.
3. $df = k - 1 -$ (Number of parameters estimated in order to find E_i)
4. If H_0 is rejected, investigate the nature if the differences using the sampling proportions.

Key Concepts

IV. Contingency Tables

1. A two-way classification with n observations into $r \times c$ cells of a two-way table using two different methods of classification is called a contingency table.
2. The test for independence of classifications methods uses the chi-square statistic

$$X^2 = \sum \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \quad \text{with} \quad \hat{E}_{ij} = \frac{r_i c_j}{n} \quad \text{and} \quad df = (r-1)(c-1)$$

3. If the null hypothesis of independence of classifications is rejected, investigate the nature of the dependency using conditional proportions within either the rows or columns of the contingency table.

Key Concepts

V. Fixing Row or Column Totals

1. When either the row or column totals are fixed, the test of independence of classifications becomes a test of the homogeneity of cell probabilities for several multinomial experiments.
2. Use the same chi-square statistic as for contingency tables.
3. The large-sample z tests for one and two binomial proportions are special cases of the chi-square statistic.

KEY CONCEPTS

VI. Assumptions

1. The cell counts satisfy the conditions of a multinomial experiment, or a set of multinomial experiments with fixed sample sizes.
2. All expected cell counts must equal or exceed five in order that the chi-square approximation is valid.